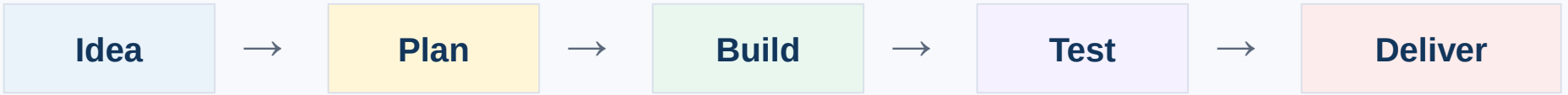
Software Project Management

Week 1

Software Projects, Life Cycle, Standards & Stakeholders

Lecture duration: 120 minutes • Semester case study: Online Shopping System



If developers write the code, why do we need a project manager?

Students usually mention deadlines, cost, team coordination, customer communication, risks, and quality. All are valid starting points.

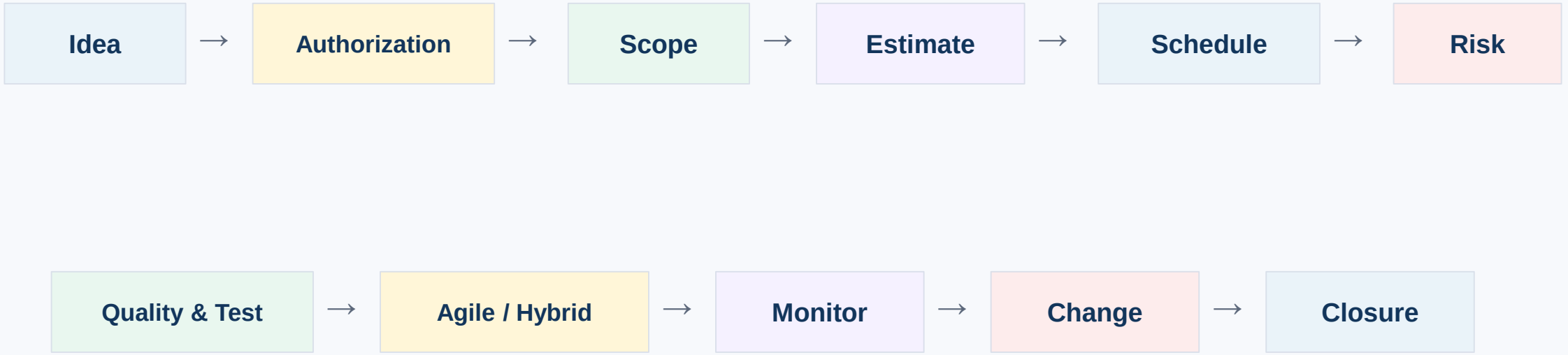
Software project management is about delivering the right software, at the required quality, within agreed time and cost constraints.

Coordination | **Cost** | **Risk** | **Quality** | **Communication** | **Delivery**

By the end of Week 1, you should be able to…

- Define a project and distinguish it from operations and products.
- Explain why software projects are difficult to manage.
- Describe the relationship between the SDLC and project management.
- Recognize predictive, iterative, incremental, Agile, and hybrid approaches.
- Identify software-project stakeholders and analyze their power and interest.
- Recognize the purpose of PMBOK, ISO/IEC/IEEE 12207, SWEBOK, and Scrum.

The semester follows a software project from idea to closure



What is a project?

A project is a temporary endeavor undertaken to create a unique product, service, or result.

Temporary
Defined beginning and end

Unique
Creates something specific

Example: Developing a new Student Advising Mobile Application is a project because it has a target release and a unique deliverable.

A software project in context

Zewail City wants a Student Advising Mobile App.

- Student login and profile.
- Course advising and recommended registration plan.
- Advisor communication and appointment requests.
- Notifications, reports, and management dashboard.
- Target release: 1 March; initial budget: 1.5 million EGP.

Project objective
Deliver an operational app with agreed features, quality, schedule, and budget.

Defined start

Defined end

Project vs. operation

Both are important, but they are managed differently.

Project

- Temporary
- Unique goal
- Defined end
- Change-oriented
- Example: develop Moodle feature

Operation

- Continuous
- Repetitive work
- Ongoing
- Stability-oriented
- Example: operate Moodle every day

Key message: developing the system is a project; running the system after release is an operation.

Project vs. product

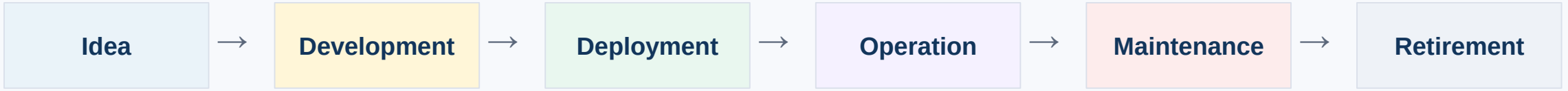
A project may end while the software product continues for years.

Project
Temporary effort to create or change something

Product
The resulting software system used after delivery

Example “Develop an Online Shopping System” is the project. The e-commerce application used by customers after launch is the product.

Product life continues after project closure



Project life

Product life

Project management must prepare the product for handover, support, and future change.

Project or operation?

- Developing a new banking mobile app. Project
- Daily monitoring of banking servers. Operation
- Migrating a database to the cloud. Project
- Processing customer support tickets. Operation
- Implementing an ERP system. Project

Why are software projects difficult to manage?

- Invisible progress
- Changing requirements
- Estimation uncertainty
- Complex dependencies
- People-intensive work
- Quality is hard to see early

These challenges explain why software teams need structured planning, monitoring, risk management, and communication.

Software progress is invisible

A building can be visually inspected; code progress is harder to judge.

When a developer says “the system is 80% complete”, what does that actually mean?

- Lines of code written?
- Features coded but not tested?
- Screens designed but not integrated?
- Core module done but payment not connected?
- Documentation and deployment still missing?

Management response: define deliverables, milestones, baselines, acceptance criteria, and measurable status reports.

Requirements change

Software projects are exposed to business, user, and technology changes.

Original requirement
Customers can pay using credit cards.

New request
Add wallets, installments, and cash-on-delivery.

Three months later…

Impact areas

- **Scope**
- **Cost**
- **Schedule**
- **Architecture**
- **Testing**

Estimation is uncertain

Managers still need early estimates to make decisions.

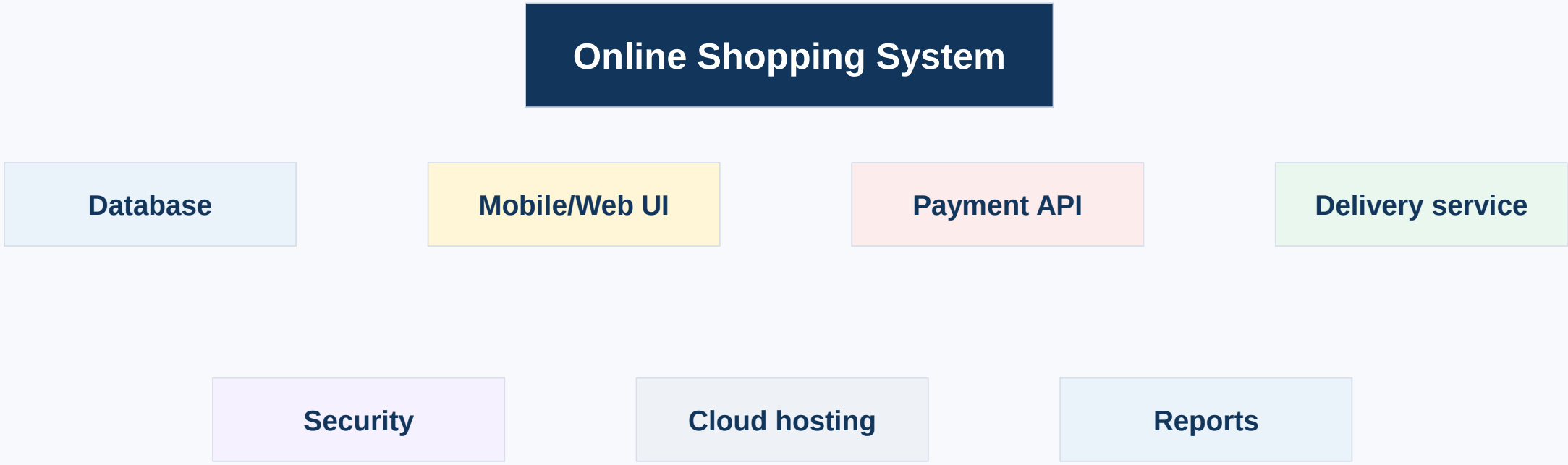
“How long will it take?”

“Maybe two months.”

- Incomplete requirements
- Unfamiliar technologies
- Different developer productivity
- Integration problems
- Defects and rework
- User feedback and changes

Week 4 will transform estimation from guessing into a structured management activity.

Modern software has many dependencies



Each dependency can introduce schedule, cost, quality, and risk issues.

Software projects are people-intensive

Adding people does not always make software faster.

Two programmers are not automatically twice as productive as one programmer.

- New members need onboarding.
- Communication paths increase.
- Some tasks cannot be split easily.
- Knowledge is unevenly distributed.
- Coordination consumes time.

Management implication
Plan resources carefully; do not assume effort and duration are the same.

Software project constraints

Project managers balance competing constraints rather than optimizing one alone.

Scope
What will be delivered?

Time
When will it be delivered?

Cost
What resources and budget?

Quality
How good is acceptable?

If scope increases while time and resources remain fixed, quality or risk is usually affected.

There is no free change.

What is the Software Development Life Cycle?

The SDLC is a structured set of activities used to develop, deliver, maintain, and eventually retire software.

- Requirements
- Analysis
- Design
- Development
- Testing
- Deployment
- Maintenance

Online Shopping System through the SDLC

Requirements	What does the business and customer need?
Design	How will screens, database, modules, and integrations work?
Development	Build catalog, cart, payment, admin, and notifications.
Testing	Verify features, security, integration, and acceptance criteria.
Deployment	Release to production and prepare users and support.

SDLC vs. Project Management Life Cycle

They interact, but they answer different questions.

SDLC asks
How do we develop the software?

Project Management asks
How do we plan, organize, monitor, control, and deliver the project?

Software engineering activities and project-management activities happen together throughout the project.

For every software activity, the manager asks…

Software activity: Develop the payment module

- Who will develop it?
- When will it start and finish?
- How long will it take?
- What will it cost?
- What dependencies exist?
- What risks exist?
- When will it be tested?
- What happens if the payment provider is late?

Different software projects need different management approaches

Predictive
Plan early; control against baseline.

Iterative
Improve through repeated cycles.

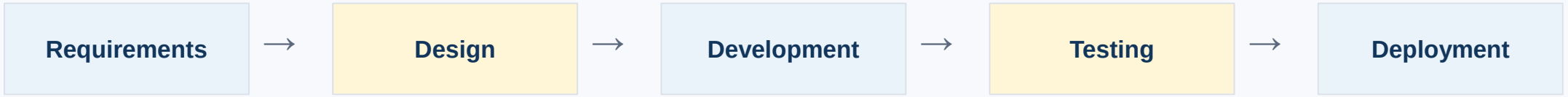
Incremental
Deliver the product in pieces.

Agile
Short iterations and feedback.

Hybrid approaches combine predictive governance with iterative or Agile delivery.

Predictive / Waterfall planning

Useful when requirements are relatively stable and approvals are formal.



- Requires early scope definition.
- Uses baselines for schedule and cost control.
- Works well when change is controlled and documentation is important.

Iterative and incremental delivery

Iteration improves; increment adds usable product functionality.

Iterative

Version 1 → Feedback → Version 2 → Feedback → Version 3

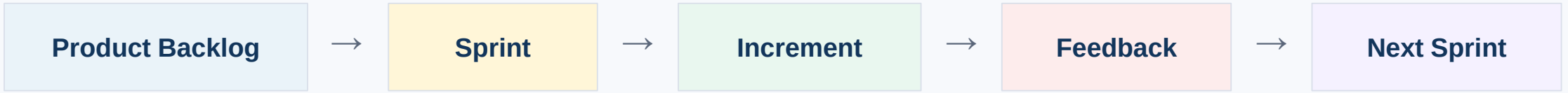
Incremental

Login + Catalog
then Cart
then Payment
then Tracking

The project manager must plan feedback cycles, release points, and dependencies between increments.

Agile development

Small iterations, frequent delivery, customer feedback, and response to change.



- Plans short horizons in detail.
- Uses backlog prioritization instead of full upfront task prediction.
- Measures progress through delivered increments, velocity, and sprint outcomes.

Hybrid management is common in real organizations

Predictive governance can coexist with Agile delivery.

Management view
Budget, major milestones, contracts, reporting

Development view
Backlog, sprints, increments, feedback

Example: A six-month approved project may be delivered internally through multiple two-week sprints.

What is a stakeholder?

A stakeholder is a person, group, or organization that can affect, be affected by, or perceive itself to be affected by the project.

- Stakeholders are not only users.
- Some stakeholders provide resources or authority.
- Some create constraints or risks.
- Some may resist the change.

Online Shopping System stakeholders

Sponsor	Project Manager	Product Owner
Sales Manager	Developers	Testers
UI/UX Designer	DBA	Security Team
Payment Provider	Delivery Company	End Customers

Good stakeholder analysis reduces communication problems, conflict, late approvals, and missed expectations.

Power–Interest Matrix

A simple way to decide stakeholder engagement strategy.

Keep Satisfied
High power / low interest

Manage Closely
High power / high interest

POWER

Monitor
Low power / low interest

Keep Informed
Low power / high interest

INTEREST

Project roles in a software project

- Sponsor: provides authority, funding support, and high-level decisions.
- Project Manager: plans, coordinates, monitors, controls, communicates, and manages risk/change.
- Product Owner / Business Owner: prioritizes business needs and clarifies value.
- Development Team: analyzes, designs, builds, tests, integrates, and prepares delivery.
- Users and external providers: influence requirements, acceptance, dependencies, and risks.

Standards and guides used in this course

We use standards to organize professional thinking, not to memorize definitions.

PMBOK

General project-management principles, performance domains, planning, governance, stakeholders, resources, and risk.

SWEBOK

Software engineering body of knowledge, including software engineering management.

ISO/IEC/IEEE 12207

Software life-cycle processes across development, operation, maintenance, and related activities.

Scrum Guide

The official Scrum framework: roles/accountabilities, events, artifacts, and commitments.

Summary + quick formative check

You now have the foundation for the rest of the semester:

Project	Product	SDLC	Management	Approach	Stakeholders

Quick questions:

- Which is a project: daily backup or developing a new payroll system?
- Can a product continue after a project is closed?
- Who normally provides high-level project authorization?
- What is the difference between SDLC and project management?

Next: Project Authorization, Business Case & Feasibility